

Explainer scripts — all 35 subtopics

KS2 maths app. 7 September 2026. English here; the French of every line is in the CSV.

These were generated from the app's own teaching data, not written alongside it. Every situation, step, reason and trap below is the exact text the subtopic already carries, so a script and the screen it explains cannot drift apart. If you change a line in the app, regenerate rather than re-editing here.

70 videos — 35 subtopics in English and French. A French child cannot use an English explainer, and the French of every line is already written; the CSV beside this has both, one row per spoken line, ready for a bulk import.

Timecodes are estimates at 135 words a minute, which is Synthesia's default slowed one notch for nine to eleven year olds. The app shows these as jump links under the player, so a child stuck on step four taps step four and the video starts there. **Send me the real marks after rendering and I will replace the estimates.**

Every one is under 90 seconds except long division, which runs to 1:52. That one has seven genuine steps and I would not cut it — but it is the one to watch in the edit.

Ten subtopics already have a drawing in the app (marked ♦ below). The film's screen text should match those exactly — same pizza, same number line, same columns — or the child is learning two different things.

One avatar for all 70. A different presenter per topic makes it a playlist; the same one makes it a teacher. Burn the captions in.

Missing angles · angles ♦

6G4a / 6G4b · target 1:01 · on screen: $130^\circ + ? = 180^\circ$ · files angles-en.mp4 / angles-fr.mp4

0:00 — the situation

A door swings open. How far did it turn?

0:05 — why it matters

Three facts cover almost every angle question: a line is 180° , a point is 360° , a triangle is 180° .

0:15 — into the method

Here is how to do it.

0:19 — step 1 — a straight line is 180°

Angles sitting on a straight line always add to 180. Half a full turn. If you can picture the turn, you never have to remember the number.

0:32 — step 2 — $180 - 130 = 50$

Subtract what you have from what it should be. Every missing-angle question is a subtraction once you know which total to use.

0:43 — step 3 — 50°

And it should look like an acute angle. Check the picture agrees. If your answer is 150° but the angle looks tiny, you have used the wrong total.

0:57 — close

That is the whole method. Try one yourself.

Area of rectangles · areaRect ♦

5M7b · target 0:53 · on screen: 8 m $\times$ 5 m · files areaRect-en.mp4 / areaRect-fr.mp4

0:00 — the situation

A garden 8 metres long and 5 metres wide. How much turf?

0:07 — why it matters

Picture it as rows of one-metre squares: 5 rows of 8.

0:13 — into the method

Here is how to do it.

0:16 — step 1 — 5 rows of 8 squares

Area is just counting squares, arranged in rows. That is why it is length times width — it is a multiplication table drawn out.

0:28 — step 2 — $8 \times 5 = 40$

Forty squares. Do not add. $8 + 5 = 13$ is the perimeter's half, not the area.

0:37 — step 3 — 40 m^2

Square metres, because you counted squares. The little 2 is not decoration. An answer in m instead of m^2 loses the mark.

0:48 — close

That is the whole method. Try one yourself.

Triangles and parallelograms · areaTri

6M7b · target 0:53 · on screen: base 10, height 6 · files areaTri-en.mp4 / areaTri-fr.mp4

0:00 — the situation

A triangular flag: base 10 cm, height 6 cm.

0:05 — why it matters

Draw a rectangle round it. The triangle is exactly half.

0:11 — into the method

Here is how to do it.

0:15 — step 1 — rectangle: $10 \times 6 = 60$

First find the rectangle it fits inside. Every triangle is half of some rectangle. That is where the formula comes from.

0:25 — step 2 — $\div 2 = 30$

Halve it. Forgetting to halve is the whole of what goes wrong here.

0:32 — step 3 — 30 cm^2

Thirty square centimetres. Height means straight up, not the slanted side. That is the other trap.

0:41 — the trap

One last thing — the mistake to watch for. Careful: don't forget to halve it.

0:48 — close

That is the whole method. Try one yourself.

Adding in columns · colAdd

3C2 / 4C2 · target 0:58 · on screen: $2\,456 + 1\,837$ · files colAdd-en.mp4 / colAdd-fr.mp4

0:00 — the situation

Two market stalls counted their takings: 2 456 and 1 837.

0:06 — why it matters

Too big to do in your head. Line them up so ones sit under ones.

0:14 — into the method

Here is how to do it.

0:18 — step 1 — $6 + 7 = 13$, write 3 carry 1

Start at the right. 13 is too big for one column. Only one digit fits in each column. The ten goes next door, where tens live.

0:31 — step 2 — $5 + 3 + 1 = 9$

Now the tens, and do not forget the one you carried. The carried digit is a real ten. Leaving it out loses ten from the answer.

0:43 — step 3 — $4 + 8 = 12$, $2 + 1 = 3 \rightarrow 4\ 293$

Carry on left. The answer is 4 293. Check it roughly: 2500 and 1800 is about 4300. It is.

0:53 — close

That is the whole method. Try one yourself.

Taking away in columns · colSub

4C2 / 5C2 · target 1:00 · on screen: $5\ 042 - 1\ 968$ · files colSub-en.mp4 / colSub-fr.mp4

0:00 — the situation

A library had 5 042 books. 1 968 were borrowed.

0:06 — why it matters

2 take away 8 does not work. You have to fetch ten from next door.

0:14 — into the method

Here is how to do it.

0:17 — step 1 — $2 - 8?$ borrow $\rightarrow 12 - 8 = 4$

The ones column is too small, so take ten from the tens. Borrowing does not change the number, it changes how it is written.

0:29 — step 2 — tens were 4, now 3: $3 - 6?$ borrow again

The tens are short too, so borrow from the hundreds — which are zero, so keep going left. A zero cannot lend. You have to reach past it to the first column that can.

0:46 — step 3 — $5\ 042 - 1\ 968 = 3\ 074$

Finish the columns. 3 074 books left. Check by adding back: $3\ 074 + 1\ 968$ should be 5 042.

0:56 — close

That is the whole method. Try one yourself.

Adding decimals · decAdd

5F10 · target 0:56 · on screen: $1.45 + 2.30$ · files decAdd-en.mp4 / decAdd-fr.mp4

0:00 — the situation

A drink at £1.45 and a sandwich at £2.30.

0:05 — why it matters

Line up the points, not the ends. That is the whole rule.

0:12 — into the method

Here is how to do it.

0:16 — step 1 — point under point

Write them with the decimal points in a vertical line. The point is where the whole numbers stop. Line that up and every column is right.

0:28 — step 2 — fill gaps with zeros

If one has fewer digits, put zeros in the empty places. 3.5 is 3.50. The zero changes nothing except making the columns match.

0:40 — step 3 — $= 3.75$

Add as normal, and keep the point in line. The point in the answer sits directly under the others. Never move it.

0:51 — close

That is the whole method. Try one yourself.

Multiplying decimals · decMul

6F9b · target 0:54 · on screen: 1.4×6 · files decMul-en.mp4 / decMul-fr.mp4

0:00 — the situation

Petrol is £1.40 a litre. You buy 6 litres.

0:05 — why it matters

Cover the point. Do 14×6 . Then put the point back where it belongs.

0:13 — into the method

Here is how to do it.

0:17 — step 1 — ignore the point: $14 \times 6 = 84$

Pretend it is a whole number for a moment. Whole-number multiplication is something you can already do. Use it.

0:27 — step 2 — one decimal place in, one out

1.4 had one digit after the point, so the answer has one. Count the places you started with. That is exactly how many you end with.

0:39 — step 3 — = 8.4

£8.40 of petrol. Check roughly: 1.4 is nearly 1.5, and $1.5 \times 6 = 9$. Close enough to trust.

0:49 — close

That is the whole method. Try one yourself.

What is left over · divRem

5C7b · target 1:03 · on screen: $29 \div 4$ · files divRem-en.mp4 / divRem-fr.mp4

0:00 — the situation

29 children need taxis. Each taxi takes 4.

0:05 — why it matters

7 taxis carry 28. One child is still standing on the pavement.

0:11 — into the method

Here is how to do it.

0:15 — step 1 — $4 \times 7 = 28$

How many whole 4s fit into 29? Seven. Whole ones only. You cannot send seven and a bit of a taxi.

0:26 — step 2 — $29 - 28 = 1$

One is left over. The remainder must be smaller than 4. If it were 4, another taxi would go.

0:35 — step 3 — $29 \div 4 = 7 \text{ r } 1$

Seven remainder one — but in real life you order 8 taxis. The maths and the answer are not always the same thing. Read the question.

0:48 — the trap

One last thing — the mistake to watch for. Careful: the remainder must be smaller than what you divided by.

0:58 — close

That is the whole method. Try one yourself.

Factors · factors ♦

5C5a / 6C5 · target 0:52 · on screen: factors of 24 · files factors-en.mp4 / factors-fr.mp4

0:00 — the situation

24 tiles. Can you lay them in a perfect rectangle?

0:06 — why it matters

1×24 , 2×12 , 3×8 , 4×6 . Every rectangle you can make gives you two factors.

0:13 — into the method

Here is how to do it.

0:16 — step 1 — 1×24

Start at 1. Every number has 1 and itself. Working in pairs means you never miss one and never repeat one.

0:27 — step 2 — 2×12 , 3×8 , 4×6

Try 2, then 3, then 4. Each that works gives a pair. Stop when the pair crosses over. After 4×6 comes 6×4 , which you already have.

0:40 — step 3 — 1 2 3 4 6 8 12 24

Eight factors. 5 is missing because 24 tiles will not make a rectangle five wide.

0:48 — close

That is the whole method. Try one yourself.

The same thing three ways · fdp

6F11 / 5F12 · target 0:58 · on screen: $1/2 = 0.5 = 50\%$ · files fdp-en.mp4 / fdp-fr.mp4

0:00 — the situation

A shirt is half price. Another is 50% off. Another is 0.5 of the price.

0:08 — why it matters

All three shops are saying exactly the same thing in three different costumes.

0:15 — into the method

Here is how to do it.

0:19 — step 1 — $1/2 \rightarrow 0.5$

Divide the top by the bottom: $1 \div 2 = 0.5$. A fraction is a division waiting to happen. That is all the line means.

0:31 — step 2 — $0.5 \rightarrow 50\%$

Multiply by 100. Per cent means out of a hundred. Decimal to percentage is always $\times 100$. Two steps left, then a % sign.

0:43 — step 3 — learn the common ones

$1/4$, $1/5$, $3/4$, $1/10$ — know these without thinking. There are about eight worth memorising. They come up in every paper.

0:53 — close

That is the whole method. Try one yourself.

Simple formulae · formula

6A2 · target 0:55 · on screen: $y = 3x + 5$, $x = 4$ · files formula-en.mp4 / formula-fr.mp4

0:00 — the situation

A recipe: cook a chicken for 40 minutes per kilogram, plus 20 minutes.

0:07 — why it matters

In maths that is written $t = 40w + 20$. The letter is just a box waiting for a number.

0:17 — into the method

Here is how to do it.

0:21 — step 1 — put 4 where x is

Swap the letter for the number you were given. $3x$ means 3 times x. The multiplication sign is hidden, not missing.

0:31 — step 2 — $3 \times 4 = 12$

Multiply first, because $\times$ comes before $+$. The order of operations still applies. It always applies.

0:40 — step 3 — $12 + 5 = 17$

So y is 17. Algebra is arithmetic with a gap in it. Fill the gap and it is ordinary sums.

0:50 — close

That is the whole method. Try one yourself.

Different bottoms · frAddDiff ♦

5F4 / 6F4 · target 1:19 · on screen: $1/2 + 1/8$ · files frAddDiff-en.mp4 / frAddDiff-fr.mp4

0:00 — the situation

Two loaves of bread, the same size. Amina cuts hers into 2 and takes 1 piece. Musa cuts his into 8 and takes 1.

0:12 — why it matters

Amina has $1/2$ and Musa has $1/8$. Put them together and you cannot say “2 pieces” — the pieces are different sizes.

0:23 — into the method

Here is how to do it.

0:27 — step 1 — $2 \times 4 = 8$

8 is a number 2 goes into. So make both bottoms 8. You are not changing the amount, only the size of the pieces you are counting in.

0:40 — step 2 — $1/2 = 4/8$

Multiply top and bottom by 4. Half a loaf is four eighths. Top and bottom by the SAME number, or you have changed how much bread there is.

0:54 — step 3 — $4/8 + 1/8 = 5/8$

Now add the tops. Five eighths. The bottom stays 8 — you are counting eighths, and adding does not change what you are counting in.

1:06 — the trap

One last thing — the mistake to watch for. Careful: never add the bottom numbers.

1:14 — close

That is the whole method. Try one yourself.

Same bottom number · frAddSame ♦

4F4 / 5F4 · target 1:02 · on screen: $3/8 + 2/8$ · files frAddSame-en.mp4 / frAddSame-fr.mp4

0:00 — the situation

A pizza cut into 8. You eat 3 slices, your brother eats 2.

0:07 — why it matters

3 slices and 2 slices is 5 slices. The slices did not change size.

0:14 — into the method

Here is how to do it.

0:18 — step 1 — $3 + 2 = 5$

Add the top numbers. The top counts how many slices. That is the only thing changing.

0:27 — step 2 — bottom stays 8

The bottom says what size the slices are. It does not change. $8 + 8 = 16$ would mean the slices got smaller, and nobody cut anything.

0:40 — step 3 — $= 5/8$

Five eighths of the pizza is gone. Adding bottoms is the commonest fraction mistake there is. Never do it.

0:49 — the trap

One last thing — the mistake to watch for. Careful: the bottom number never changes.

0:57 — close

That is the whole method. Try one yourself.

Dividing by a whole number · frDivWhole

6F5b · target 0:51 · on screen: $1/3 \div 2$ · files frDivWhole-en.mp4 / frDivWhole-fr.mp4

0:00 — the situation

A third of a pie, shared between two people.

0:05 — why it matters

Each gets half of a third. Cut every third in two and you have sixths.

0:13 — into the method

Here is how to do it.

0:17 — step 1 — each piece halves

Sharing between 2 makes every piece half the size. You are not making fewer pieces, you are making smaller ones.

0:27 — step 2 — $3 \times 2 = 6$

Twice as many pieces in the whole pie, so the bottom doubles. Dividing by 2 makes the bottom bigger. That feels backwards until you see the pie.

0:40 — step 3 — $= 1/6$

One sixth each. Check: two sixths make a third. It works.

0:46 — close

That is the whole method. Try one yourself.

Multiplying fractions · frMul

6F5a · target 0:54 · on screen: $1/3 \times 1/2$ · files frMul-en.mp4 / frMul-fr.mp4

0:00 — the situation

Half a chocolate bar. You give a third of that half away.

0:07 — why it matters

A third of a half is a sixth. “Of” means multiply, and the answer gets smaller.

0:15 — into the method

Here is how to do it.

0:19 — step 1 — tops: $1 \times 1 = 1$

Multiply the top numbers together. No common bottom needed. This is the easiest of the four and it looks the hardest.

0:29 — step 2 — bottoms: $3 \times 2 = 6$

Multiply the bottom numbers together. The pieces get smaller, so the bottom number gets bigger. That is the sense of it.

0:40 — step 3 — $= 1/6$

One sixth of the bar. Multiplying two fractions always makes something smaller than both. That is your check.

0:49 — close

That is the whole method. Try one yourself.

A fraction of an amount · frOfAmount

5F5 / F1 · target 0:55 · on screen: $3/4$ of 60 · files frOfAmount-en.mp4 / frOfAmount-fr.mp4

0:00 — the situation

£60 in your pocket. You spend three quarters of it.

0:06 — why it matters

Split the 60 into 4 equal parts first. Then take three of them.

0:13 — into the method

Here is how to do it.

0:16 — step 1 — $60 \div 4 = 15$

Divide by the bottom number. That is one quarter. The bottom tells you how many pieces to cut the amount into.

0:27 — step 2 — $15 \times 3 = 45$

Multiply by the top number. You want three of those quarters. Bottom first, then top. Doing it the other way round still works but the numbers get ugly.

0:41 — step 3 — = £45

Forty-five pounds spent. Three quarters should be most of it but not all. 45 out of 60 looks right.

0:50 — close

That is the whole method. Try one yourself.

Long division · longDiv

6C7b · target 1:52 · on screen: $486 \div 18$ · files longDiv-en.mp4 / longDiv-fr.mp4

0:00 — the situation

Fatou has 486 mangoes to take to the market. She packs them into boxes, 18 to a box.

0:09 — why it matters

She needs to know how many boxes to bring — too few and the mangoes spoil, too many and she is carrying air.

0:21 — into the method

Here is how to do it.

0:24 — step 1 — $18 \overline{) 486}$

Write it the long way. The 486 goes inside, the 18 outside. The layout is doing the remembering for you — it keeps track of what is left.

0:38 — step 2 — 18 into 4? no

Can 18 go into 4? No. So look at 48 instead. You always start with the smallest part of the number that is big enough.

0:50 — step 3 — 18 into 48 = 2, $2 \times 18 = 36$

18 twos are 36. 18 threes are 54 — too big. So 2. You are asking how many whole 18s fit, never more.

1:02 — step 4 — $48 - 36 = 12$

Take the 36 away. 12 is left over from the 48. That 12 has not been shared yet, so it must come down into the next step.

1:15 — step 5 — bring down the 6 → 126

Bring the 6 down beside the 12. Now you have 126. The 12 was really 120. Adding the 6 makes 126, which is what is genuinely left.

1:28 — step 6 — 18 into 126 = 7, $7 \times 18 = 126$

18 sevens are exactly 126. Nothing left. When it lands exactly, there is no remainder — every mango is in a box.

1:39 — step 7 — $486 \div 18 = 27$

Fatou needs 27 boxes. Check it the other way: 27×18 should give 486 back.

1:48 — close

That is the whole method. Try one yourself.

Long multiplication · longMul

5C7a / 6C7a · target 1:25 · on screen: 46×27 · files longMul-en.mp4 / longMul-fr.mp4

0:00 — the situation

A builder is laying a wall. Each row takes 46 bricks, and the wall is 27 rows high.

0:09 — why it matters

He has to order the bricks before he starts. Order too few and the lorry comes back next week.

0:19 — into the method

Here is how to do it.

0:23 — step 1 — $46 \times 7 = 322$

First the 7 rows. 46 sevens are 322. Splitting 27 into 7 and 20 is the whole trick — both parts are easy on their own.

0:35 — step 2 — $46 \times 20 = 920$

Now the 20 rows. $46 \times 2 = 92$, so $46 \times 20 = 920$. THIS is why the second row starts with a zero. You are multiplying by twenty, not by two.

0:51 — step 3 — $322 + 920 = 1242$

Add the two rows together. 7 rows plus 20 rows is 27 rows. The addition is the reassembling.

1:01 — step 4 — $46 \times 27 = 1\,242$

He orders 1 242 bricks. The forgotten zero is the commonest mistake in this method, and it costs the whole mark.

1:11 — the trap

One last thing — the mistake to watch for. Careful: the second row starts with a zero.

1:20 — close

That is the whole method. Try one yourself.

The mean · mean

6S3 · target 0:50 · on screen: 5, 3, 10 · files mean-en.mp4 / mean-fr.mp4

0:00 — the situation

Three fish weighed 5 kg, 3 kg and 10 kg.

0:06 — why it matters

The mean is what each would weigh if you shared the weight out evenly.

0:13 — into the method

Here is how to do it.

0:17 — step 1 — $5 + 3 + 10 = 18$

Add them all together. You are pooling everything before sharing it back out.

0:24 — step 2 — $18 \div 3 = 6$

Divide by how many there were. Count how many, not how big. Three fish means divide by three.

0:33 — step 3 — mean = 6 kg

Six kilograms each, if they were all the same. The mean sits between the smallest and the biggest. If yours does not, check again.

0:45 — close

That is the whole method. Try one yourself.

Using what you know · mentalPV

0:00 — the situation

Binta has 30 boxes of eggs. Seven eggs in every box.

0:06 — why it matters

She already knows 3 sevens are 21. Thirty boxes is ten times three boxes.

0:14 — into the method

Here is how to do it.

0:17 — step 1 — $3 \times 7 = 21$

Do the easy one first. Three sevens are 21. You are not learning a new fact. You are using one you have.

0:28 — step 2 — 30 is 10 lots of 3

Thirty boxes is ten times as many as three boxes. Ten times bigger going in means ten times bigger coming out.

0:39 — step 3 — $21 \times 10 = 210$

So 30×7 is 210. This is the most-asked thing in the whole test. It is worth doing until it is automatic.

0:50 — close

That is the whole method. Try one yourself.

Changing units · metric

0:00 — the situation

A bottle says 1.5 litres. The recipe asks for millilitres.

0:06 — why it matters

Milli means a thousandth. So one litre is a thousand of them.

0:12 — into the method

Here is how to do it.

0:16 — step 1 — $1 \text{ l} = 1000 \text{ ml}$

Know the four that matter: km-m, m-cm, kg-g, l-ml. Three of those four are $\times 1000$. Only m to cm is $\times 100$.

0:27 — step 2 — smaller unit $\rightarrow$ bigger number

Going to a smaller unit means more of them, so multiply. Ask yourself: is the answer going to be bigger or smaller? Then you cannot pick the wrong way.

0:42 — step 3 — $1.5 \times 1000 = 1500 \text{ ml}$

Fifteen hundred millilitres. 1.5 litres could never be 0.0015 of anything. The size check catches the slip.

0:50 — the trap

One last thing — the mistake to watch for. Careful: smaller unit means a bigger number.

0:59 — close

That is the whole method. Try one yourself.

Which comes first · ooo

0:00 — the situation

Tickets cost £4. You buy 3 and you already had £5.

0:06 — why it matters

$5 + 3 \times 4$. Left to right gives 32. That is wrong — and it is wrong by a lot.

0:17 — into the method

Here is how to do it.

0:20 — step 1 — $3 \times 4 = 12$

Multiplication happens first, wherever it sits. $\times$ and $\div$ bind more tightly than $+$ and $-$. They grab their numbers first.

0:31 — step 2 — $5 + 12 = 17$

Then the addition. 17, not 32. Reading left to right is the single most expensive habit in arithmetic.

0:40 — step 3 — brackets beat everything

$(5 + 3) \times 4 = 32$. Brackets say “do me first”. If you want the adding first, you have to say so with brackets.

0:53 — the trap

One last thing — the mistake to watch for. Careful: not left to right.

1:00 — close

That is the whole method. Try one yourself.

Percentages of amounts · pcOf

6R2 · target 1:06 · on screen: 15% of £60 · files pcOf-en.mp4 / pcOf-fr.mp4

0:00 — the situation

A pair of trainers costs £60. The shop takes 15% off in the sale.

0:07 — why it matters

Per cent means “out of a hundred”. 15% means 15 for every 100.

0:14 — into the method

Here is how to do it.

0:18 — step 1 — 10% of 60 = 6

10% is one tenth. Divide by 10. 10% is the one you should never have to think about. Everything else is built from it.

0:30 — step 2 — 5% of 60 = 3

5% is half of 10%. Half of 6 is 3. Halving is easier than dividing by 20, and it is the same thing.

0:42 — step 3 — $6 + 3 = 9$

15% is 10% and 5% together. £9 off. Any percentage can be built from 10%, 5% and 1%. You never need a formula.

0:53 — the trap

One last thing — the mistake to watch for. Careful: find 10% first, then build it.

1:01 — close

That is the whole method. Try one yourself.

Ten, a hundred, a thousand · pow10 ♦

5C6b / 6F9a · target 1:06 · on screen: 24×10 · files pow10-en.mp4 / pow10-fr.mp4

0:00 — the situation

One balloon costs 24p. How much for ten? For a hundred?

0:06 — why it matters

The digits take a step to the left each time. They do not gain a zero — they move.

0:16 — into the method

Here is how to do it.

0:20 — step 1 — $24 \rightarrow 240$

Every digit shifts one place left. The 4 was ones, now it is tens. The zero appears because the ones column is now empty, not because you added it.

0:34 — step 2 — $\times 100 =$ two places

A hundred is ten tens, so two steps left. One zero in the 10, two in the 100, three in the 1000 — that is how many steps.

0:47 — step 3 — $\div 10 =$ one place right

Dividing sends them back the other way. This is why 3.5×10 is 35 and not 3.50. Adding a zero is a trick that breaks on decimals.

1:01 — close

That is the whole method. Try one yourself.

Prime numbers · primes

5C5c / 6C5 · target 1:02 · on screen: is 13 prime? · files primes-en.mp4 / primes-fr.mp4

0:00 — the situation

13 biscuits. Can you share them fairly between any group?

0:06 — why it matters

Not between 2, 3, 4, 5, 6... only between 1 person or 13. That makes 13 prime.

0:14 — into the method

Here is how to do it.

0:18 — step 1 — 2? no. 3? no. 4? no.

Try dividing by each small number. You only need to try up to the point where the pairs would cross over.

0:29 — step 2 — only 1 and 13 work

Exactly two factors: itself and one. Exactly two. Not one, not three. That is the definition.

0:37 — step 3 — 1 is NOT prime

1 has only one factor, so it fails the test. This catches people every year. 1 is not prime and 2 is — the only even one.

0:50 — the trap

One last thing — the mistake to watch for. Careful: 1 is not prime.

0:58 — close

That is the whole method. Try one yourself.

What a digit is worth · pvDigit ♦

5N3a · target 1:01 · on screen: 352 741 · files pvDigit-en.mp4 / pvDigit-fr.mp4

0:00 — the situation

A stadium says 352 741 people came this season. Someone asks: is that a lot of 5s?

0:09 — why it matters

There is one 5 in that number. But it is not worth five. It is worth fifty thousand.

0:18 — into the method

Here is how to do it.

0:22 — step 1 — 3 5 2 7 4 1

Count the places from the right: ones, tens, hundreds, thousands, ten thousands, hundred thousands. Always from the right. The right-hand end is where counting starts.

0:34 — step 2 — 5 is in the ten thousands

The 5 is the fifth place along, which is ten thousands. Each step left multiplies by ten. That is the whole idea of place value.

0:46 — step 3 — $5 \times 10\,000 = 50\,000$

So the 5 is worth 50 000. Same digit, different place, different worth. Move it and the number changes completely.

0:57 — close

That is the whole method. Try one yourself.

Below zero · pvNeg ♦

4N5 / 5N5 · target 1:07 · on screen: $-5 \rightarrow 3$ · files pvNeg-en.mp4 / pvNeg-fr.mp4

0:00 — the situation

It was 3 degrees at breakfast. By night it had dropped to 5 below zero.

0:08 — why it matters

How far did it fall? Not 2. It fell 3 to reach zero, then 5 more. Eight.

0:17 — into the method

Here is how to do it.

0:20 — step 1 — count up to zero: 5

From -5 up to 0 is five steps. Zero is the meeting point. Always go there first and the sign stops confusing you.

0:32 — step 2 — then on to 3: 3

From 0 up to 3 is three more. The second half is ordinary counting. The minus sign was only ever in the first half.

0:44 — step 3 — $5 + 3 = 8$

Add the two halves: the gap is 8. Never subtract the two numbers. -5 and 3 are 8 apart, not 2.

0:54 — the trap

One last thing — the mistake to watch for. Careful: count up to zero, then on.

1:03 — close

That is the whole method. Try one yourself.

Rounding · pvRound

4N4b / 5N4 · target 1:05 · on screen: $3\,472 \rightarrow$ nearest 1000 · files pvRound-en.mp4 / pvRound-fr.mp4

0:00 — the situation

A school trip needs coaches. 187 children are going, and each coach holds 50.

0:07 — why it matters

Nobody says “about 187”. They say “about 200”. Rounding is saying a number the way people actually say it.

0:17 — into the method

Here is how to do it.

0:21 — step 1 — look at the hundreds: 4

To round to the nearest thousand, look at the digit one place to the right. That digit is the whole decision. Nothing to the right of it matters at all.

0:35 — step 2 — 4 is less than 5 $\rightarrow$ stay

4 is less than 5, so the thousands stay as they are. It is nearer 3000 than 4000, and that is all rounding means: which is it closer to.

0:50 — step 3 — $3\,472 \rightarrow 3\,000$

Everything after the thousands becomes zero. You are not deleting the digits, you are saying the number to the nearest thousand.

1:00 — close

That is the whole method. Try one yourself.

Sharing unequally · ratioShare ♦

6R4 · target 1:09 · on screen: 30 in the ratio 3 : 2 · files ratioShare-en.mp4 / ratioShare-fr.mp4

0:00 — the situation

30 sweets shared between two brothers, 3 for one and 2 for the other each time.

0:08 — why it matters

Count the shares first. 3 and 2 makes 5 shares, not 2.

0:15 — into the method

Here is how to do it.

0:19 — step 1 — $3 + 2 = 5$ shares

Add the numbers in the ratio to find how many shares there are. This step is the one everybody skips, and skipping it makes every answer wrong.

0:32 — step 2 — $30 \div 5 = 6$

Divide the total by the number of shares. One share is 6. Now you know what a single share is worth, and the rest is multiplying.

0:45 — step 3 — $3 \times 6 = 18$ and $2 \times 6 = 12$

18 and 12. They add back to 30. Always add your two answers back. If they do not make the total, something is wrong.

0:57 — the trap

One last thing — the mistake to watch for. Careful: count the shares first.

1:04 — close

That is the whole method. Try one yourself.

Short division · shortDiv

5C7b · target 1:00 · on screen: $852 \div 4$ · files shortDiv-en.mp4 / shortDiv-fr.mp4

0:00 — the situation

852 sweets shared fairly between 4 children.

0:04 — why it matters

Work left to right this time — the opposite of adding. Share the biggest pieces first.

0:13 — into the method

Here is how to do it.

0:16 — step 1 — $8 \div 4 = 2$

Start at the left. 8 hundreds shared by 4 is 2 hundreds each. Division starts at the big end because that is how you actually share things out.

0:30 — step 2 — $5 \div 4 = 1$ r 1, carry the 1

4 goes into 5 once, and 1 is left over. Carry it to the next digit. The leftover 1 is really 10, which is why it joins the next column.

0:45 — step 3 — $12 \div 4 = 3 \rightarrow 213$

The carried 1 makes 12. $12 \div 4$ is 3. Each child gets 213. Check: 213×4 should give 852 back.

0:56 — close

That is the whole method. Try one yourself.

Short multiplication · shortMul

4C7 · target 0:51 · on screen: 248×6 · files shortMul-en.mp4 / shortMul-fr.mp4

0:00 — the situation

A shop sells juice in crates of 6. There are 248 crates.

0:07 — why it matters

Break 248 into its columns and multiply each one, right to left.

0:13 — into the method

Here is how to do it.

0:17 — step 1 — $8 \times 6 = 48$, write 8 carry 4

Ones first. 48 is four tens and eight. Same rule as adding: one digit per column, the rest moves left.

0:27 — step 2 — $4 \times 6 = 24$, $+4 = 28$

Tens next. Multiply, then add what you carried. Multiply first, add the carry after. Doing it the other way gives the wrong answer.

0:38 — step 3 — $2 \times 6 = 12$, $+2 = 14 \rightarrow 1\ 488$

1 488 bottles. Roughly $250 \times 6 = 1500$, so 1 488 is believable.

0:46 — close

That is the whole method. Try one yourself.

Squares and cubes · sqcube

5C5d · target 1:06 · on screen: 7^2 and 4^3 · files sqcube-en.mp4 / sqcube-fr.mp4

0:00 — the situation

A square patio, 7 slabs along each side. How many slabs?

0:06 — why it matters

7 rows of 7 is 49. That is what “seven squared” means — a square.

0:14 — into the method

Here is how to do it.

0:18 — step 1 — $7^2 = 7 \times 7 = 49$

The little 2 says how many sevens to multiply together. It never means $\times 2$. 7^2 is 49, not 14 — that is the commonest mistake here.

0:31 — step 2 — $4^3 = 4 \times 4 \times 4$

Cubed means three of them, like a cube with sides of 4. Squared is a flat square, cubed is a solid box. The words describe shapes.

0:44 — step 3 — $= 64$

$16 \times 4 = 64$. Do it in steps. 4×4 first, then $\times 4$ again — not all at once.

0:53 — the trap

One last thing — the mistake to watch for. Careful: 7^2 is 49, not 14.

1:01 — close

That is the whole method. Try one yourself.

Tables to twelve · tables

4C6a · target 0:50 · on screen: 6×7 · files tables-en.mp4 / tables-fr.mp4

0:00 — the situation

Every morning Binta collects eggs. Six nests, seven eggs in each.

0:06 — why it matters

She could count them one by one, every morning, for the rest of her life. Or she could know that 6 sevens are 42.

0:18 — into the method

Here is how to do it.

0:22 — step 1 — $6 \times 7 = 42$

Six sevens are forty-two. This one is worth knowing cold — it turns up inside long division constantly.

0:31 — step 2 — $7 \times 6 = 42$

And the other way round is the same. You only ever learn half a table. 6 rows of 7 and 7 rows of 6 are the same eggs, counted differently.

0:46 — close

That is the whole method. Try one yourself.

Two unknowns · twoUnknowns

6A4 · target 0:52 · on screen: $a + b = 7$, $a \times b = 12$ · files twoUnknowns-en.mp4 / twoUnknowns-fr.mp4

0:00 — the situation

Two numbers. They add to 7. They multiply to 12.

0:06 — why it matters

Do not guess wildly. List the pairs that multiply to 12 and check which adds to 7.

0:14 — into the method

Here is how to do it.

0:18 — step 1 — pairs making 12: 1×12 , 2×6 , 3×4

Start with the multiplying, because there are fewer options. Multiplication narrows things down fast. Addition has far too many pairs.

0:28 — step 2 — which adds to 7? $3 + 4$

Check each pair against the second fact. Two facts about two numbers is always enough. You just have to use both.

0:39 — step 3 — 3 and 4

The numbers are 3 and 4. Test them both ways before you write anything down.

0:47 — close

That is the whole method. Try one yourself.

Volume · volume ♦

6M8a · target 0:48 · on screen: $4 \times 3 \times 5$ · files volume-en.mp4 / volume-fr.mp4

0:00 — the situation

A box 4 cm by 3 cm by 5 cm. How many centimetre cubes fit?

0:08 — why it matters

Fill the bottom layer first, then count how many layers.

0:14 — into the method

Here is how to do it.

0:17 — step 1 — bottom layer: $4 \times 3 = 12$

Twelve cubes cover the base. The base is just an area. You already know how to do that bit.

0:27 — step 2 — 5 layers

The box is 5 cubes tall. Volume is area with a third dimension stacked on top of it.

0:36 — step 3 — $12 \times 5 = 60 \text{ cm}^3$

Sixty cubic centimetres. Little 3 this time, because you counted cubes rather than squares.

0:44 — close

That is the whole method. Try one yourself.